Functions & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Function transformations

Let $f(x) = x^2$ and $g(x) = -3f(x+2) - 4 = -3(x+2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.

2 [Foundational] • Target: Function transformations

Using the same $g(x) = -3(x+2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g's graph.

3 [Foundational] • Target: Even and odd functions

Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.

4 [Foundational] • Target: Even and odd functions

Explain why $k(x) = x^2 + x$ is neither even nor odd.

5 [Foundational] • Target: Domain restrictions

Find the domain of $f(x) = \sqrt{2x+6}$.

6 [Foundational] • Target: Even and odd functions

Consider the function $f(x) = \sqrt{2x-5} + 3$. (a) Find the domain and range of $f(x)$. (b) Find the explicit equation for the inverse function $f^{-1}(x)$, including its restricted domain.

7 [Foundational] • Target: Domain restrictions

Let $f(x) = \frac{2x+1}{x-3}$ for $xeq3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise functions

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2+5x-3}{x^2+4x+3}$.

9 [Intermediate] • Target: Function transformations

Given $f(x) = e^x$ and $g(x) = \ln(x+1)$, find the domain of $(g \circ f)(x)$ and evaluate $(g \circ f)(0)$.

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10 **[Intermediate]** • **Target: Even and odd functions**

The function $f(x) = \frac{4x-7}{2x+8}$ has horizontal asymptote $y = H$ and vertical asymptote $x = V$. Determine the value of $H + V$.

11 **[Intermediate]** • **Target: Domain restrictions**

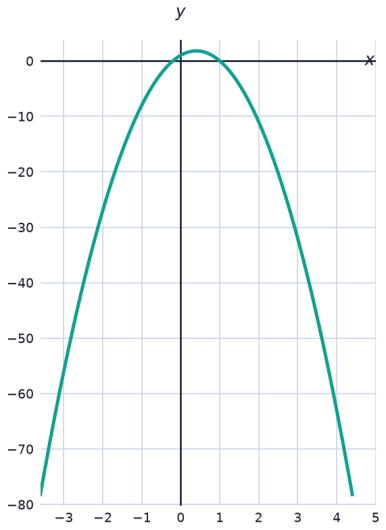
The graph of $y = f(x)$ is transformed to $y = -2f(x-3) + 5$. Describe the sequence of geometric transformations applied to the parent function.

12 **[Intermediate]** • **Target: Piecewise functions**

If $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x+1}$, find the domain of the composite function $(g \circ f)(x)$.

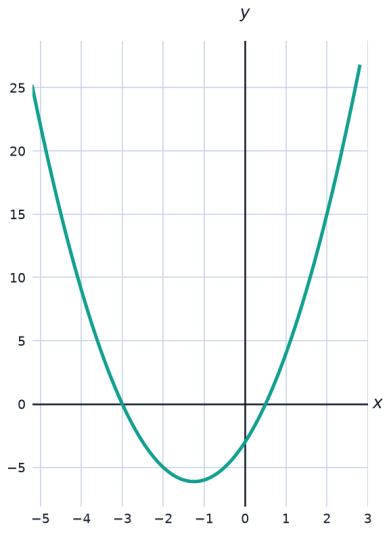
13 **[Intermediate]** • **Target: Function transformations**

Find the slant (oblique) asymptote of the rational function: $g(x) = \frac{2x^3 - 5x^2 + 4x + 1}{x^2 - x + 1}$



14 **[Intermediate]** • **Target: Even and odd functions**

Find the vertical and horizontal asymptotes of the rational function: $f(x) = \frac{4x^2 - 1}{2x^2 + 5x - 3}$



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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Domain restrictions** • **Cross-Lesson: 2.01 Sequences & Series**

The polynomial $P(x) = 2x^3 + kx^2 - 7x + 6$ has $(x - 2)$ as a factor. (a) Find the value of k. (b) Factor $P(x)$ completely over the real numbers and list all its roots.

16 **[Challenging]** • **Target: Piecewise functions** • **Cross-Lesson: 4.01 Matrices & Vectors**

Solve the logarithmic equation for x: $\log_3(x + 2) + \log_3(x - 4) = 3$

17 **[Challenging]** • **Target: Function transformations** • **Cross-Lesson: 5.01 Introductory Calculus**

Solve the rational equation for all real values of x: $\frac{x}{x-2} + \frac{1}{x-4} = \frac{2}{(x-2)(x-4)}$

18 **[Challenging]** • **Target: Even and odd functions** • **Cross-Lesson: 6.01 Probability**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2+1}{x^2-4}$. Describe its end behavior using limits.

19 **[Challenging]** • **Target: Domain restrictions** • **Cross-Lesson: 1.01 Complex Numbers**

Determine the value of the constant k for which the function $f(x)$ is continuous at $x = 2$:

$f(x) = \{kx^2 + 2x, , x \leq 2; 3x + k, , x > 2$

20 **[Challenging]** • **Target: Piecewise functions** • **Cross-Lesson: 2.01 Sequences & Series**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Function transformations**

If f is an odd function and $f(2) = 13$, find $f(-2)$.

a. -13

b. 13

c. -12

d. 2

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22 [Foundational] • Target: Even and odd functions

If f is an even function and $f(3) = -7$, find $f(-3)$.

a. -7

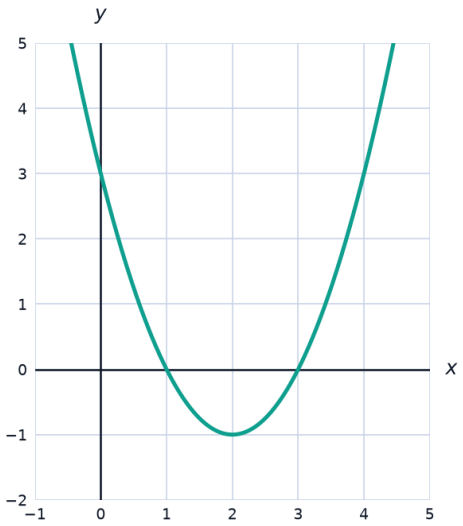
b. 7

c. -6

d. 3

23 [Foundational] • Target: Domain restrictions

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?



a. (2, -1)

b. (2, 1)

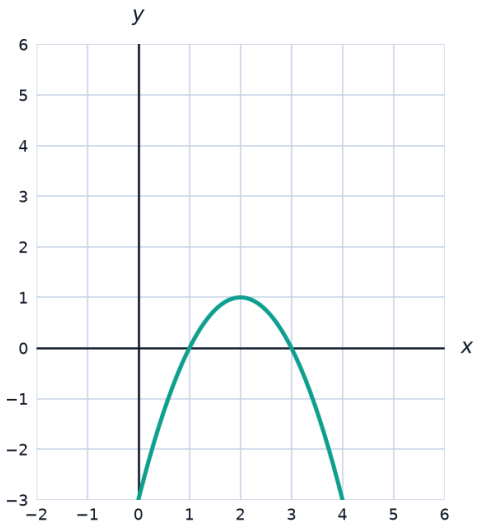
c. (1, 0)

d. (3, 0)

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24 [Foundational] • Target: Piecewise functions

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.



a. 1

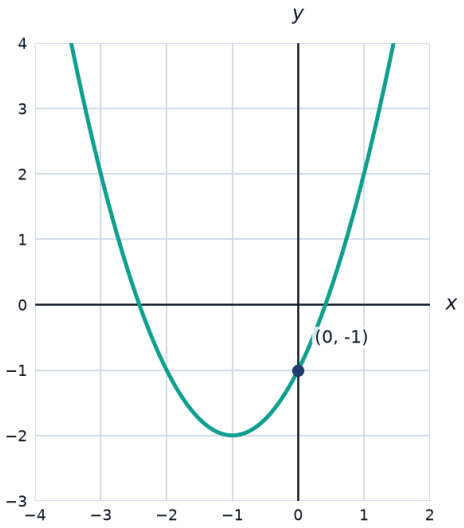
b. 2

c. 3

d. 4

25 [Foundational] • Target: Function transformations

Consider the graph of the function $f(x)$ shown below. What is the value of $f(0)$?



a. -1

b. 1

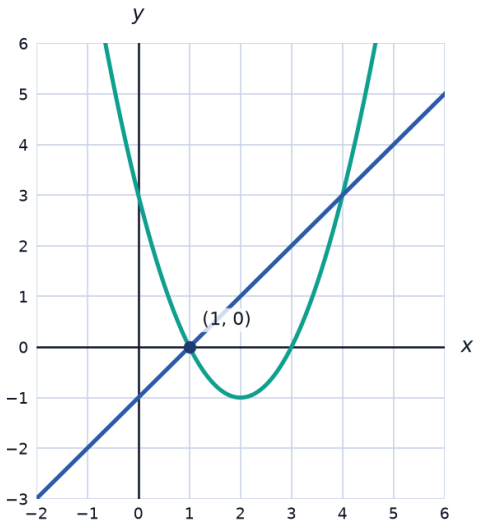
c. 0

d. 2

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26 [Foundational] • Target: Even and odd functions

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?



a. 0

b. 2

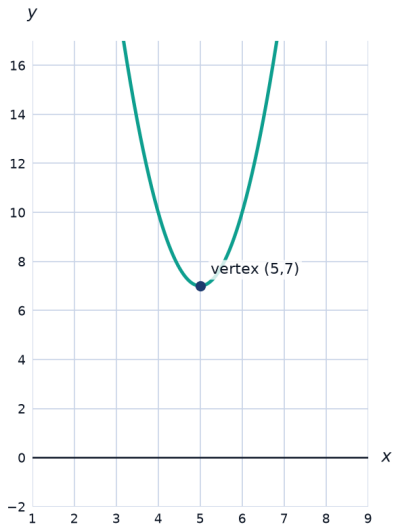
c. -1

d. 3

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27 [Foundational] • Target: Domain restrictions

Describe the transformation from $f(x)$ to $g(x) = 3f(x - 5) + 7$.



a. Shift right 5, stretch by 3, shift up 7.

b. Shift left 5, stretch by 3, shift up 7.

c. Shift right 5, stretch by 3, shift down 7.

d. Shift right 5, shift up 7 (no stretch).

28 [Foundational] • Target: Piecewise functions

The graph of $f(x) = x^2$ is shifted to $g(x) = (x + 6)^2 - 4$. Find the vertex of g.

a. (-6,\ -4)

b. (6,\ -4)

c. (-6,\ 4)

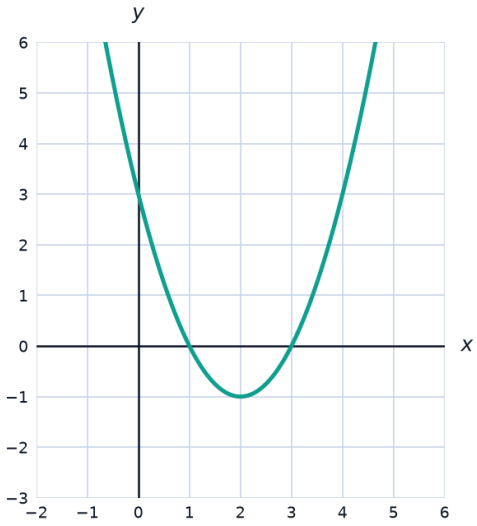
d. (-4,\ -6)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Function transformations**

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. What is the y-intercept of this parabola?



a. 3

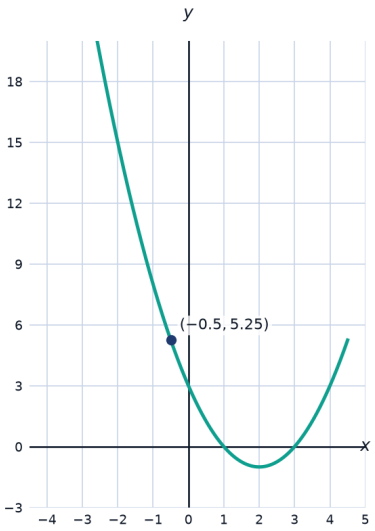
b. -3

c. 0

d. -4

30 **[Intermediate]** • **Target: Even and odd functions**

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Find the x-coordinate of the vertex of the parabola.



a. 1

b. 2

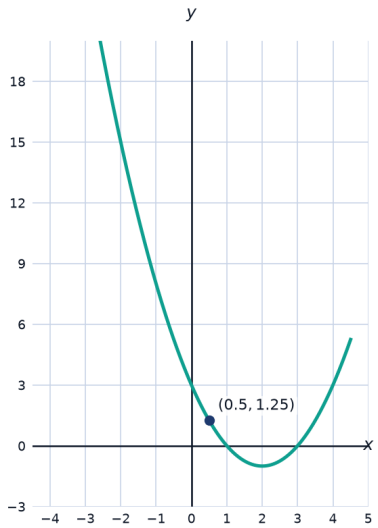
c. 3

d. -1

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31 **[Intermediate]** • **Target: Domain restrictions**

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



a. (1, 0) and (3, 0)

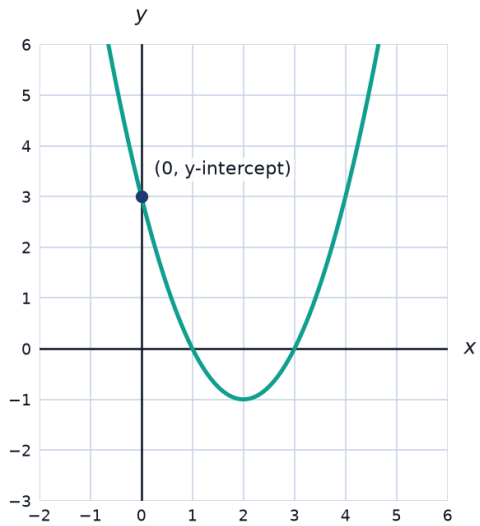
b. (0, 1) and (0, 3)

c. (-1, 0) and (-3, 0)

d. (1, 0) and (-3, 0)

32 **[Intermediate]** • **Target: Piecewise functions**

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?



a. 3

b. -3

c. 4

d. -4

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33 [Intermediate] • Target: Function transformations

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. Odd

c. Neither

d. Cannot be determined

34 [Intermediate] • Target: Even and odd functions

Consider the function $f(x) = x^3 - 6x^2 + 5$. A tangent line to the curve of $f(x)$ is drawn at the point $P(a, f(a))$. This tangent line also intersects the curve at another point $Q(b, f(b))$. If the x-coordinate of the point P is $a = 1$, determine the x-coordinate of the point Q.

a. 6

b. 5

c. 4

d. 3

35 [Intermediate] • Target: Domain restrictions

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation $f(x) + f(\frac{x-1}{x}) = 1 + x$. What is the value of $f(2)$?

a. $\frac{3}{4}$

b. $\frac{5}{4}$

c. $\frac{7}{4}$

d. $\frac{9}{4}$

36 [Intermediate] • Target: Piecewise functions

If $f(x) = \sqrt{x-5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \leq x \leq 3$

b. $x \leq -3$ or $x \geq 3$

c. $x \geq 3$

d. All real numbers

37 [Intermediate] • Target: Function transformations

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. -5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Even and odd functions • Cross-Lesson: 6.01 Probability

A piecewise function is defined by $f(x) = 2x + 3$ for $x < 1$ and $f(x) = x + k$ for $x \geq 1$. Find k so that f is continuous at $x = 1$.

a. 5

b. 6

c. 4

d. 1

39 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

A survey was conducted on a group of 100 students regarding their preferences for two subjects, Mathematics (M) and Science (S). The results are summarized in the two-way table below, where some values are unknown. | | Science (S) | Not Science (¬S) | Total | |-------------|-------------|------------------|-------| | Mathematics (M) | 25 | x | 45 | | Not Math (¬M) | y | 30 | 55 | | Total | 55 | z | 100 | If a student is chosen at random from this group, the probability that the student prefers Mathematics but not Science, P(M ∩ ¬S), is equal to the probability that the student prefers Science but not Mathematics, P(¬M ∩ S). What is the value of x?

a. 15

b. 20

c. 25

d. 30

40 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

A transformation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation can be represented by matrix multiplication $(x\ y)' = M(x\ y) + (a\ b)$. Find the values of a and b.

a. a = -5, b = 0

b. a = 5, b = 0

c. a = 0, b = -5

d. a = 0, b = 5

41 [Challenging] • Target: Function transformations • Cross-Lesson: 4.01 Matrices & Vectors

A linear transformation in the 2D plane is represented by the matrix $M = (a, b; c, d)$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.

a. 9

b. 11

c. 7

d. 5

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42 [Challenging] • Target: Even and odd functions • Cross-Lesson: 5.01 Introductory Calculus

A manufacturer produces light bulbs. The probability that a randomly selected light bulb is defective is p. The production process is designed such that the probability of producing exactly k defective bulbs in a batch of $n = 5$ bulbs follows the binomial probability formula: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. It is known that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Find the value of p.

a. 0.2

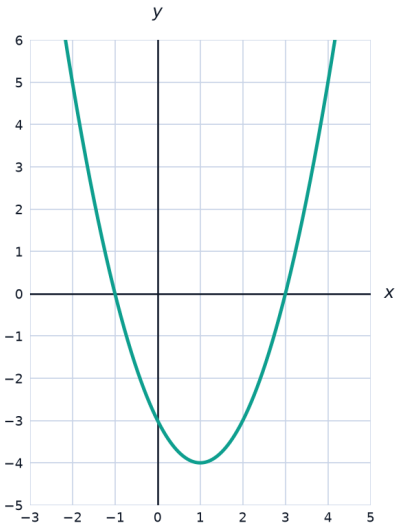
b. 0.5

c. 0.8

d. 0.75

43 [Challenging] • Target: Domain restrictions • Cross-Lesson: 6.01 Probability

The graph of the function $f(x) = (x - 1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?



a. $[-4, \infty)$

b. $[1, \infty)$

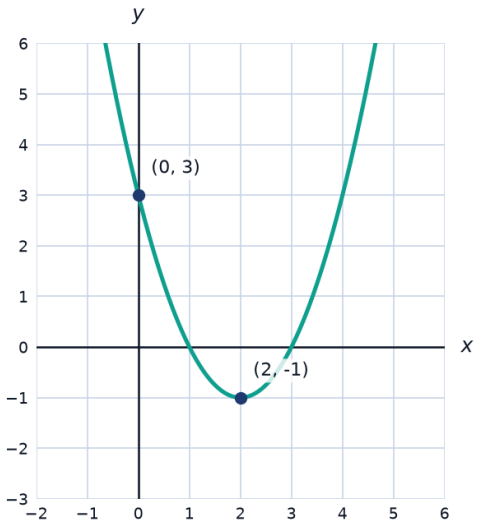
c. $(-\infty, -4]$

d. $(-\infty, \infty)$

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44 [Challenging] • Target: Piecewise functions • Cross-Lesson: 1.01 Complex Numbers

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$. The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?



a. -5

b. -3

c. 1

d. 5

45 [Challenging] • Target: Function transformations • Cross-Lesson: 2.01 Sequences & Series

Consider the transformation grid below. If point A has coordinates $(1, 2)$, what are the coordinates of its image, A', after the transformation shown?

a. (-4, 2)

b. (-2, 1)

c. (2, -4)

d. (1, -2)

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